



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2010

HSC Mathematics Task 2

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Start each **NEW SECTION** in a separate answer booklet.
- Hand in three separate sections labelled A, B and C.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.

Total Marks – 72

- Attempt all questions 1-6.
- All answers are to be given in simplest exact form unless specified.

Examiner: *P.Bigelow*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Section A

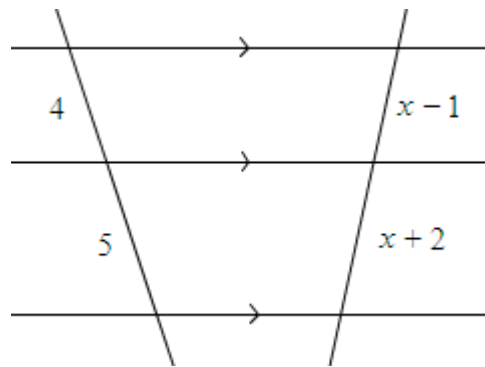
Question One [12 marks]

(a) Find a primitive of $3 + x^2$. 1

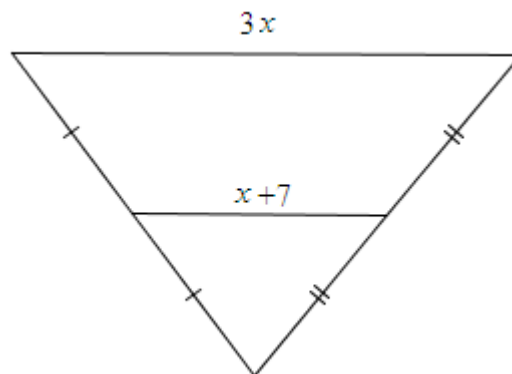
(b) Find the value of $\log_e 6$ correct to two decimal places. 1

(c) Find the value of x in the following: 2

(i)



(ii)



(d) Find the exact value of: 2

$$\cos \frac{\pi}{4} + \sin \frac{2\pi}{3}$$

- (e) Differentiate 6
- (i) e^{-3x}
- (ii) $\ln(x+4)$
- (iii) $x \sin(x+2)$
- (iv) $\frac{x}{\cos x}$

Question Two [12 marks]

- (a) Find the equation of the tangent to the curve $y = 3 \log_e x$ at the point $(1, 0)$. 2
- (b) Find 8
- (i) $\int \sec^2 3x dx$
- (ii) $\int \frac{4}{3+x} dx$
- (iii) $\int (1-x)^2 dx$
- (iv) $\int_0^{\ln 4} e^{2x} dx$
- (c) The graph of $y = F(x)$ passes through $(4, 1)$ and $F'(x) = \frac{2}{\sqrt{x}}$, find $F(x)$. 2

End of Section A

Section B

(Start a NEW answer booklet)

Question Three [12 marks]

(a)

4

- (i) Without using calculus draw a neat sketch of $y = 3\cos 2x$ for $-\pi \leq x \leq \pi$.
- (ii) On the same set of axes draw the line $y = x$.
- (iii) How many solutions has the equation $3\cos 2x = x$?

(b) The following table lists the values of a function for five values of x .

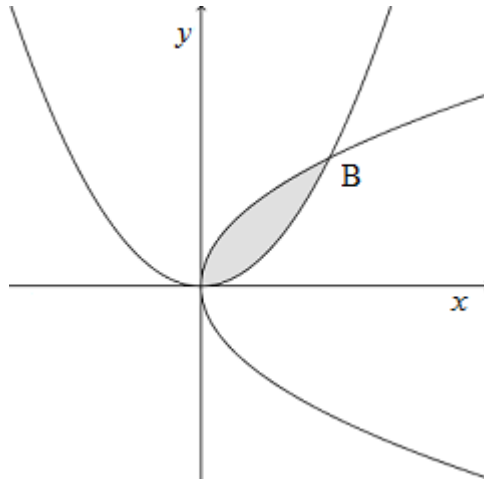
4

x	2	3	4	5	6
$f(x)$	1.6	2.9	7.6	3.2	0.8

Use these five function values to estimate $\int_2^6 f(x)dx$ by

- (i) the trapezoidal rule.
- (ii) Simpson's rule.

- (c) The curves $y = x^2$ and $y^2 = 8x$ intersect at the origin and the point B. 4



- (i) Find the coordinates of B.
- (ii) Find the area between the curves $y = x^2$ and $y^2 = 8x$.

Question Four [12 marks]

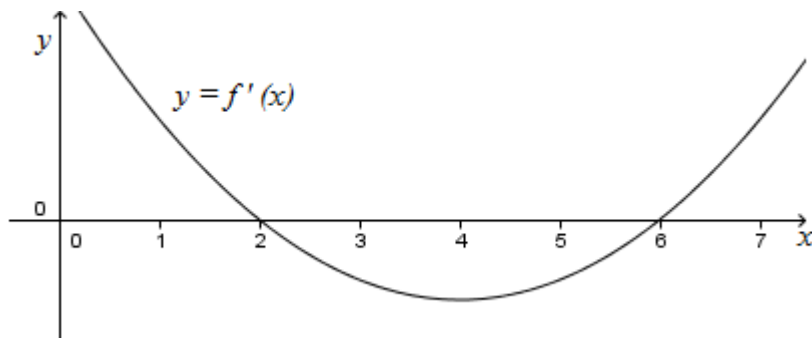
- (a) Solve $\cos \theta = -0.3$ for $0 \leq \theta \leq 2\pi$. 2
[Give your answer in radians correct to two decimal places]

- (b) Let $f(x) = 2x^3 - 3x^2 - 12x - 15$. 6

- (i) Find the coordinates of the stationary points and determine their nature.
- (ii) Find the coordinates of the point of inflexion.
- (iii) Sketch the graph of $y = f(x)$ clearly indicating the stationary points and the point of inflexion.

- (c) Show that $f(x) = \frac{1}{1 + \tan x}$ is decreasing for all values of x for which it is defined. 2

- (d) 2



The diagram shows the graph of the gradient function of the curve $y = f(x)$. For what values of x does $f(x)$ have a local minimum? Justify your answer.

End of Section B

Section C

(Start a NEW answer booklet)

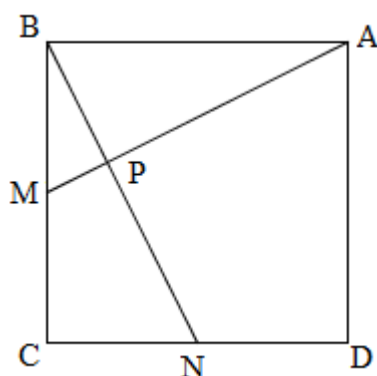
Question Five [12 marks]

- (a) Given $f(x) = \frac{1}{e^{x^2}}$, find 3

(i) $f'(x)$

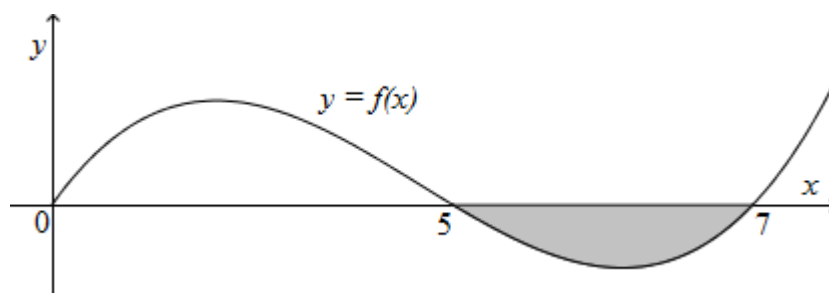
(ii) $f''(x)$

- (b) ABCD is a square, M and N are the mid-points of BC and CD respectively. 4



- (i) Prove that triangles ABM and BCN are congruent.
- (ii) Prove that AM and BN are perpendicular.

- (c) 2

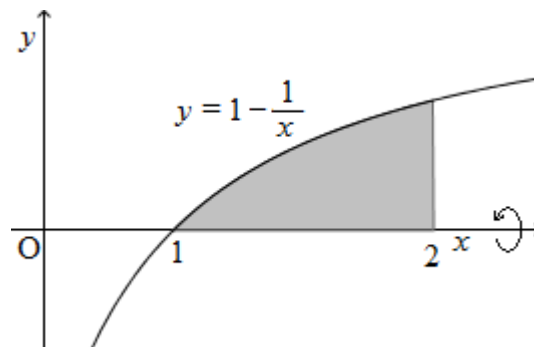


Given that $\int_0^5 f(x)dx = 20$ and $\int_0^7 f(x)dx = 14$.

- (i) What is the area of the shaded region?
- (ii) What is the value of $\int_5^7 f(x)dx$?

(d)

3

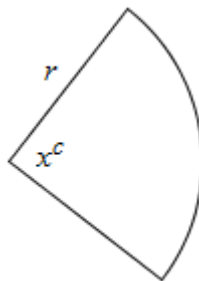


The area under the curve $y = 1 - \frac{1}{x}$ between $x = 1$ and $x = 2$ is rotated about the x -axis. Find in exact form the volume generated.

Question Six [12 marks]

(a) A sector of a circle has a perimeter of 21 cm and an area of 27 cm^2 .

5



- Use the information to form a pair of simultaneous equations.
- Solve these equations to find all values of r and x , which satisfy the information.

(b)

- (i) Find the minimum value of

3

$$x + \frac{900}{x} \text{ for } x > 0.$$

[show all working]

- (ii) A transport company operates a shipping service between port A and port B, at a constant speed of v km/h. The distance between the ports is S km.

4

For a given v , the cost *per hour* of running the ship is $\$(9000 + 10v^2)$.

Find the value of v which minimizes the cost of the trip.

End of Section C

End of Exam

2010 HSC Mathematics Task 2 – Section A:

Question One:

a) $\int 3 + x^2 \cdot dx = 3x + \frac{x^3}{3} + C$

b) $\log_e 6 = 1.79$

c)

(i)
$$\frac{x-1}{x+2} = \frac{4}{5}$$
$$5x - 5 = 4x + 8$$
$$x = 13$$

(ii)
$$x + 7 = \frac{1}{2}(3x)$$
$$2x + 14 = 3x$$
$$x = 14$$

d) $\cos \frac{\pi}{4} + \sin \frac{2\pi}{3} = \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}$

e)

(i) $\frac{d}{dx}(e^{-3x}) = -3e^{-3x}$

(ii) $\frac{d}{dx} \ln(x + 4) = \frac{1}{x+4}$

(iii) $\frac{d}{dx}[x \sin(x + 2)] = \sin(x + 2) + x \cos(x + 2)$

(iv) $\frac{d}{dx} \left[\frac{x}{\cos x} \right] = \frac{\cos x + x \sin x}{\cos^2 x}$

Question Two:

a) $y = 3 \log_e x$

$$y' = \frac{3}{x}$$

At $x = 1$,

$$y' = \frac{3}{1} = 3$$

The equation of the tangent is

$$y - 0 = 3(x - 1)$$

$$\therefore y = 3x - 3$$

b)

(i) $\int \sec^2 3x \cdot dx = \frac{1}{3} \tan 3x + C$

(ii) $\int \frac{4}{3+x} \cdot dx = 4 \ln(3+x) + C$

(iii) $\int (1-x)^2 \cdot dx = -\frac{(1-x)^3}{3} + C$

(iv)
$$\begin{aligned} \int_0^{\ln 4} e^{2x} \cdot dx &= \left[\frac{1}{2} e^{2x} \right]_0^{\ln 4} \\ &= \frac{1}{2} e^{2 \ln 4} - \frac{1}{2} e^0 \\ &= 8 - \frac{1}{2} \\ &= 7\frac{1}{2} \end{aligned}$$

c)
$$\begin{aligned} F(x) &= \int \frac{2}{\sqrt{x}} \cdot dx \\ &= 2 \int x^{-1/2} \cdot dx \\ &= 2 \times 2x^{1/2} + C \end{aligned}$$

$$F(x) = 4\sqrt{x} + C$$

At $(4, 1)$,

$$1 = 4\sqrt{4} + C$$

$$1 = 4 \times 2 + C$$

$$\therefore C = -7$$

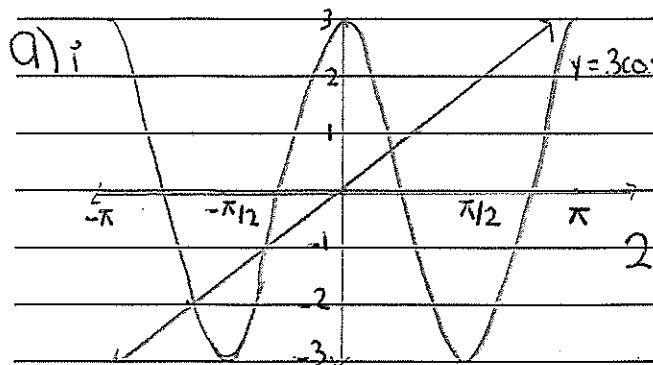
$$\therefore F(x) = 4\sqrt{x} - 7$$

Section B

$x = 2$

Question 3

when $x = 2$ $y = 2^2$
 $= 4$



$\therefore B = (2, 4)$ 2

ii $A = \int_0^2 (\sqrt{8x} - x^2) dx$

$= \int_0^2 (\sqrt{8} x^{1/2} - x^2) dx$

$= \left[\frac{\sqrt{8} x^{3/2}}{3/2} - \frac{x^3}{3} \right]_0^2$

$= \left[\frac{4\sqrt{2} x^{3/2}}{3} - \frac{x^3}{3} \right]_0^2$

$= \left[\frac{4\sqrt{2} \cdot 2^{3/2}}{3} - \frac{2^3}{3} \right] - [0 - 0]$

$= \left[\frac{4\sqrt{2} \times \sqrt{8}}{3} - \frac{8}{3} \right]$

$= \left[\frac{16}{3} - \frac{8}{3} \right]$

$= \frac{8}{3} \text{ units}^2$ 2

b) i $\int_2^6 f(x) dx$ $h = \frac{6-2}{4} = 1$

$\approx \frac{1}{2} [(1 \cdot 6 + 0 \cdot 8) + 2(2 \cdot 9 + 7 \cdot 6 + 3 \cdot 2)]$

$= 14.9$ 2

ii $\int_2^6 f(x) dx$

$\approx \frac{1}{3} [(1 \cdot 6 + 0 \cdot 8) + 4(2 \cdot 9 + 3 \cdot 2) + 2(7 \cdot 6)]$

$= 14$ 2

c) i $y = x^2$
 $y^2 = 8x$

$\therefore (x^2)^2 = 8x$

$x^4 = 8x$

$x^4 - 8x = 0$

$x(x^3 - 8) = 0$

$\therefore x = 0, x^3 - 8 = 0$
 $x^3 = 8$

Question 4

x	0	$\frac{1}{2}$	1
$f''(x)$	-6	0	+6

a) $\cos \theta = -0.3$
for $0 \leq \theta \leq 2\pi$

$\checkmark$ A $\theta = 1.88, 4.41$ (2dp)
 $\checkmark$ C

Since concavity changes
 $(\frac{1}{2}, -2\frac{1}{2})$ is a horizontal
point of inflexion. 2

b) $f(x) = 2x^3 - 3x^2 - 12x - 15$

i) $f'(x) = 6x^2 - 6x - 12$
st pts. $f'(x) = 0$

$0 = 6x^2 - 6x - 12$

$0 = x^2 - x - 2$

$0 = (x-2)(x+1)$

$\therefore x = 2, x = -1$
 $\therefore f(x) = -35, f(x) = -8$

x	1	2	3	$\therefore$ local
$f'(x)$	-12	0	24	minimum
x	-2	-1	0	$\therefore$ local
$f'(x)$	24	0	-12	maximum

c) $f(x) = (1 + \tan x)^{-1}$
 $f'(x) = -1(1 + \tan x)^{-2} \times \sec^2 x$
 $= \frac{-\sec^2 x}{(1 + \tan x)^2}$

as anything squared is
always positive you have

$f'(x) = -x \pm$

which will always be negative
 $\therefore f(x)$ is decreasing for
all values of x for
which it is defined. 2

ii) $f''(x) = 12x - 6$
 $f''(x) = 0$ for p.o.i

$0 = 12x - 6$

$\therefore 0 = 2x - 1$

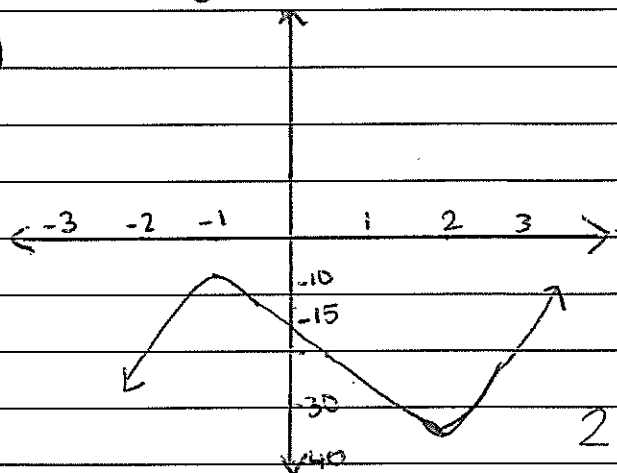
$\therefore 2x = 1$

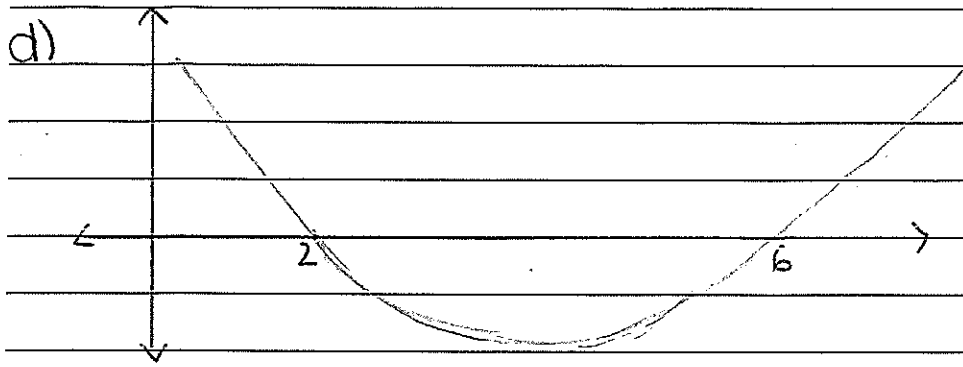
$x = \frac{1}{2}$

when $x = \frac{1}{2}$

$f(x) = -2\frac{1}{2}$

$\therefore (\frac{1}{2}, -2\frac{1}{2})$ is a possible
point of inflexion



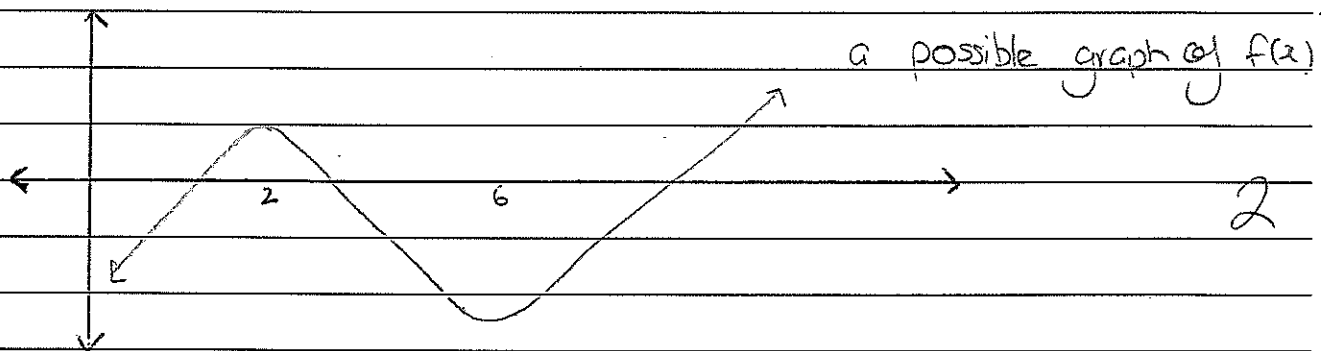


$x < 2$ gradient is positive

$x = 2, x = 6$ zero gradient $\therefore$ stationary point

$2 < x < 6$ gradient is negative

$x > 6$ gradient is positive



$\therefore f(x)$ is a local minimum when $x = 6$

OR for local minimum gradient needs to go from negative to positive ie this happens @ $x = 6$.

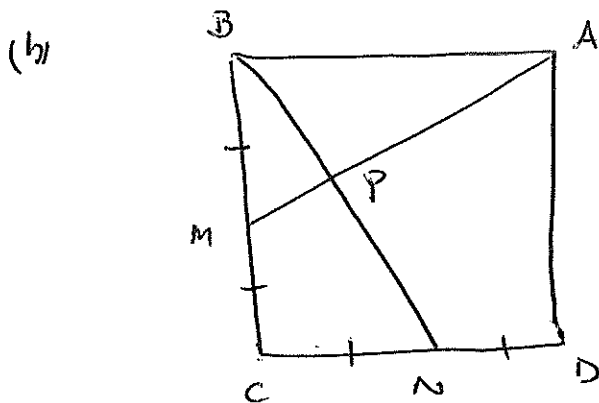
SECTION C

Question 5

$$(a) \quad p(x) = \frac{1}{e^{x^2}} \\ = e^{-x^2}$$

$$(i) \quad f'(x) = e^{-x^2} \cdot -2x = -2x e^{-x^2} \quad 1$$

$$(ii) \quad f''(x) = e^{-x^2} \cdot -2 + (-2x) \cdot (-2x) e^{-x^2} \\ = -2e^{-x^2} + 4x^2 e^{-x^2} \\ = (4x^2 - 2) e^{-x^2} \quad 2$$



$$(i) \quad AB = BC \quad (\text{sides of square})$$

$$BM = \frac{1}{2} BC = \frac{1}{2} CD = CN \quad (BC = CD \text{ sides of square}) \quad 2$$

$$\angle ABM = \angle BCN = 90^\circ \quad (\text{angles of square})$$

$$\therefore \triangle ABM \cong \triangle BCN \quad (\text{SAS})$$

$$(ii) \quad \angle PBM = \angle NBC \quad (\text{common}) \quad 2$$

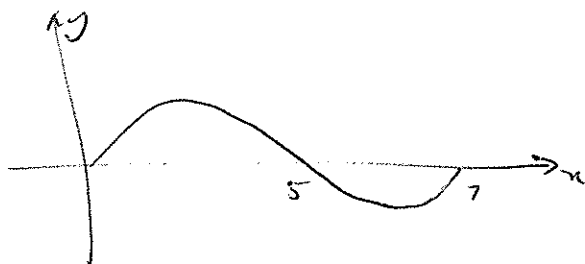
$$\angle PMB = \angle ANB \quad (\text{common})$$

$$= \angle BNC \quad (\text{corresponding } \angle \text{ in } \triangle \text{S})$$

$$\therefore \triangle BPM \cong \triangle BCN \quad (\text{equilateral})$$

$$\therefore \angle BPM = \angle BCN = 90^\circ \quad (\text{corresponding } \angle \text{ in } \triangle \text{S}) \quad \therefore AM \perp BN$$

(c)



$$\int_0^5 f(x) dx = 20 \quad \int_5^7 f(x) dx = 14$$

(i) Area = 6 units²

(ii) $\int_5^7 f(x) dx = -6$

(d) $V_1 = \pi \int_1^2 \left(1 - \frac{1}{x}\right)^2 dx$

$$= \pi \int_1^2 \left(1 - \frac{2}{x} + \frac{1}{x^2}\right) dx$$

$$= \pi \left[x - 2 \ln x - \frac{1}{x} \right]_1^2$$

$$= \pi \left\{ \left[2 - 2 \ln 2 - \frac{1}{2} \right] - \left[1 - 0 - 1 \right] \right\}$$

$$= \pi \left[\frac{3}{2} - 2 \ln 2 \right] \text{ units}^3$$

3

12

Question 6

$$(R) \quad (i) \quad P = 2r + xr = 21$$

$$A = \frac{1}{2} r^2 x = 27$$

$$\therefore x = \frac{54}{r^2}$$

$$\therefore 2r + \frac{54}{r^2} \cdot r = 21$$

$$\therefore 2r + \frac{54}{r} = 21$$

$$\therefore 2r^2 - 21r + 54 = 0$$

$$\therefore 2r^2 - 9r - 12r + 54 = 0$$

$$\therefore r(2r-9) - 6(2r-9) = 0$$

$$\therefore (2r-9)(r-6) = 0$$

$$\therefore r = 4\frac{1}{2} \quad \text{or} \quad r = 6$$

$$\therefore x = 2\frac{2}{3} \quad x = 1\frac{1}{2}$$

x	r
108	-27
2, 54	
3, 36	
4, 27	
6, 18	
9, 12	

$$(b) \quad (i) \quad V = x + \frac{900}{x}$$

$$V' = 1 - \frac{900}{x^2}$$

$$V'' = \frac{1800}{x^3}$$

For min value $V' = 0$, $V'' > 0$

$$\text{If } V' = 0, \quad 1 - \frac{900}{x^2} = 0$$

$$\therefore x^2 = 900$$

$$x = 30$$

(as $x > 0$)

when $x = 30$, $V'' > 0$

$\therefore$ min value when $x = 30$

$$\text{min value} = 30 + \frac{900}{30}$$

$$= 60$$

(ii) $T = \frac{\text{Distance}}{\text{speed}}$

$$= \frac{S}{r}$$

$$\text{Cost} = T \times (9000 + 10r^2)$$

$$= \frac{S}{r} (9000 + 10r^2)$$

$$= 10S \left(r + \frac{900}{r} \right)$$

min cost occurs when $r = 30$ (using (i))